OPTION TIME VALUE

Learning Outcome

- explain the exercise value, moneyness, and time value of an option

While the exercise value of an option reflects its *current* payoff, an additional component of an option's value is derived from its remaining *time* to maturity. Although European options can be exercised only at maturity, they can be purchased or sold prior to maturity at a price (c_t or p_t, for a call or put, respectively) that reflects the option's *future expected* payoff. A longer time until expiration usually means a higher potential dispersion of the future underlying price for a given level of volatility. Similarly, an increase in volatility at a specific underlying price for a given time to expiration increases option value for the same reason. We will explore these factors further in a later lesson.

The time value of an option is equal to the difference between the current option price and the option's current payoff (or exercise value):

Call Option Time Value: $c_t - Max\left(0,\ S_t - X(1+r)^{-(T-t)}\right)$ 3

or: $c_t = Max\left(0,\ S_t - X(1+r)^{-(T-t)}\right) + \text{Time Value}$

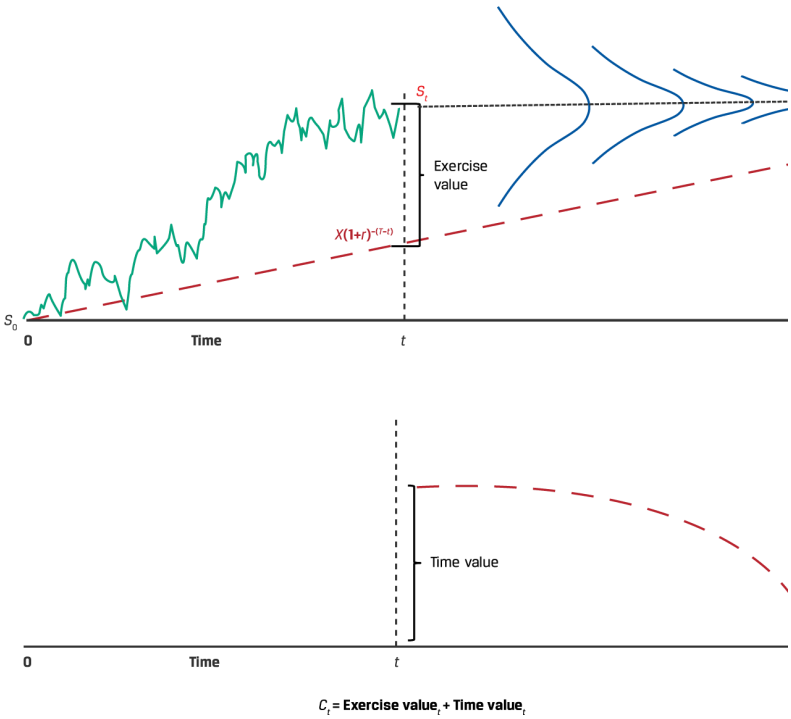
Put Option Time Value: $p_t - Max\left(0,\ X(1+r)^{-(T-t)} - S_t\right)$ 4

or: $p_t = Max\left(0,\ X(1+r)^{-(T-t)} - S_t\right) + \text{Time Value}$

That is, the current option price is equal to the sum of its exercise value and time value. As Exhibit 2 shows, the time value of an option is always positive but declines to zero at maturity, a process referred to as **time value decay**.

Exhibit 2:

Exercise Value and Time Value of a Call Option



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Pricing and Valuation of Options

Related Questions:
Practice questions related to this topic

EXAMPLE 3

Put Option Time Value

Example 1 showed that a one-year put option with an exercise price (X) of EUR 1,000 had an exercise value of EUR 45.04 with six months remaining to maturity when the spot price (S_t) was EUR 950. If we observe a current put option price (p_t) of EUR 50, what is the time value of the put option?

Use Equation 4 to solve for p_t − Max (0, PV(X) − S_t):

$$p_t - \max\left(0,\ X(1+r)^{-(T-t)} - S_t\right)$$

= EUR 4.96 (= EUR 50 − EUR 45.04)

Question Set: Option Value relative to Underlying Spot Price

1. Describe the similarities and differences between the exercise value of a long put option position with an exercise price (X) equal to the forward price, $F_0(T)$, and a short forward position payoff at maturity with the same underlying details.

   Hide Solution

   Solution:

   The exercise value of a put option with an exercise price of $X = F_0(T)$ at maturity (Max (0, $X - S_T$)) is the *same* as the payoff of a short forward commitment at maturity ($F_0(T) - S_T$) *as long as* the exercise price is greater than the underlying price. That is, for $F_0(T) = X$:

   If $X > S_T$: $F_0(T) - S_T$ = Max (0, $X - S_T$)

   These payoff profiles will *differ* if $S_T > X$, since the short forward payoff at maturity will require a payment from the seller to the buyer, while the put option owner will allow the option to expire unexercised. Note that this payoff comparison ignores the upfront option premium paid by the put option buyer (p_0) at time t = 0.

2. A European put option with three months remaining to maturity on an underlying stock with no additional cash flows has an exercise price (X) of GBP 50, a risk-free rate of 2%, and a current underlying price (S_t) of GBP 55. If the current put option price is GBP 5, calculate the exercise value and the time value and interpret the results.

   Hide Solution

   Solution:

   An option's value comprises its exercise value plus its time value. The exercise value of a put option is Max (0, PV(X) − S_t) and can be calculated as follows:

   $$\text{Put Option Exercise Value} = Max\left(0,\ X(1+r)^{-(T-t)} - S_t\right)$$

   Max(0, GBP 50$(1.02)^{-0.25}$ − GBP 55)

   = 0

   The exercise value is zero, as the current underlying price exceeds the present value of the exercise price. The time value is the difference between the option price and the exercise value and represents the possibility that the likelihood

$$\text{Put Option Time Value} = p_t - Max\left(0,\ X(1+r)^{-(T-t)} - S_t\right)$$

= GBP 5 (= GBP 5 − GBP 0)

The time value is always positive and declines to zero at maturity ($t = T$). Since the put option is out of the money, its value consists *solely* of time value.

3. Match the following underlying price and exercise price relationships with their associated call option:

1. $S_T = 60$, $X = 50$	A. An at-the-money call option
2. $S_T = 50$, $X = 50$	B. An in-the-money call option
3. $S_T = 40$, $X = 50$	C. An out-of-the-money call option

Hide Solution

Solution:

Call options are in the money when $S_T > X$, at the money when $S_T = X$, and out of the money when $S_T < X$. Therefore:

1. B is correct. Since $S_T > X$, this is an in-the-money call option.
2. A is correct. Since $S_T = X = 50$, this is an at-the-money call option.
3. C is correct. Since $S_T < X$, this is an out-of-the-money call option.

4. Describe how the moneyness of an option affects how the option's value will change for a given change in the price of the underlying.

Hide Solution

Solution:

An increase in the moneyness of an option will increase the sensitivity of its value to changes in the underlying price. For example, an option that is very likely to be exercised will have a nearly one-to-one change in option value for a given change in the underlying price, while an option that is unlikely to be exercised will have a relatively small change in value for a given change in the underlying price.

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Hi Everyone!

I suppose the equation (3) is missing the ct parameter.

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